

## Module 6: Interest and Loans — Summary

**What this module is about.** Interest connects arithmetic growth to real financial decisions. Simple interest adds the same dollar amount each period because it is always computed from the original principal. Compound interest multiplies the current balance by the same growth factor, so the dollar increase itself grows over time. Loans use the same percentage-growth idea, but a payment is subtracted after interest is added.

If  $P$  is principal,  $r$  is the annual rate as a decimal, and  $t$  is time in years, simple interest is

$$A = P(1 + rt).$$

If interest is compounded  $n$  times per year,

$$A = P \left( 1 + \frac{r}{n} \right)^{nt}.$$

Simple interest has a constant dollar increase. Compound interest has a constant multiplicative factor per compounding period.

### Example 1: Simple versus compound growth

Invest \$1500 at 6% for 10 years.

Simple interest:

$$A = 1500(1 + 0.06 \cdot 10) = 1500(1.6) = \$2400.$$

Annual compounding:

$$A = 1500(1.06)^{10} \approx \$2686.27.$$

The compound balance is larger because later interest is earned on earlier interest.

For percentage growth near ordinary financial rates, the doubling time can be estimated by

$$t \approx \frac{72}{r\%},$$

where  $r\%$  is the annual percentage rate written as a percent number. At 6%, the estimated doubling time is  $72/6 = 12$  years. To estimate a quadrupling time, remember that quadrupling is two doublings.

For a loan balance  $B$ , annual rate  $r$ , and monthly payment  $M$ , the monthly rate is

$$i = \frac{r}{12}.$$

During one month:

$$\text{interest} = iB, \quad \text{principal reduction} = M - iB,$$

$$B_{\text{new}} = B(1 + i) - M.$$

If the payment is not larger than the interest charge, the balance will not decrease.

**Example 2: One month of a loan**

A \$3000 loan has 9% APR compounded monthly and a \$150 payment.

$$i = \frac{0.09}{12} = 0.0075.$$

First-month interest:

$$0.0075(3000) = \$22.50.$$

Of the \$150 payment,  $150 - 22.50 = \$127.50$  reduces principal. The new balance is

$$3000 + 22.50 - 150 = \$2872.50.$$

A larger payment on the same starting balance does not change the first month's interest; it changes how much principal is removed and therefore changes future interest.

**By the end of this module, you should be able to:** calculate simple and compound balances; identify the number of compounding periods and the periodic rate; use the Rule of 72; and perform a one-period loan update showing interest, principal reduction, and new balance.

## Module 6: Interest and Loans — Prequiz

*Open-note. Show your mathematical work. A calculation and a clearly stated final answer are enough unless a sentence is specifically requested.*

Name:

Date:

### Useful formulas

$$A = P(1 + rt), \quad A = P \left( 1 + \frac{r}{n} \right)^{nt}, \quad t_{\text{double}} \approx \frac{72}{r\%},$$

$$B_{\text{new}} = B \left( 1 + \frac{r}{12} \right) - M.$$

1. A student deposits \$1500 in an account that earns 4% simple interest per year for 3 years.
  - (a) Find the total amount in the account after 3 years.
  - (b) How many dollars of interest are added each year under this simple-interest model?
  
2. A savings account starts with \$1000 and earns 5% compounded annually for 4 years.
  - (a) Write the compound-interest expression.
  - (b) Find the balance to the nearest cent.
  - (c) Explain in one short phrase why the dollar amount of interest is not the same every year.
  
3. Use the Rule of 72 for an investment growing at approximately 6% per year.
  - (a) Estimate the doubling time.
  - (b) Estimate the time required to grow to roughly four times the original amount.
  
4. A \$3000 loan charges 9% APR compounded monthly. The borrower pays \$150 at the end of each month.
  - (a) Find the monthly rate as a decimal.
  - (b) Find the interest charged during the first month.
  - (c) How much of the first payment reduces principal?
  - (d) Find the balance immediately after the first payment.

5. Two loans have the same starting balance and the same interest rate. Loan A has a monthly payment of \$150, while Loan B has a monthly payment of \$250. Which loan has the larger first-month interest charge? Which loan will generally be paid off sooner? Explain each answer using the loan update process rather than guessing from the payment size alone.

## Module 6: Interest and Loans — Matching Quiz

*These problems use the same skills as the prequiz, with new values and contexts. Show enough work to make your reasoning visible.*

Name:

Date:

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1. A \$2000 account earns 3% simple interest for 5 years. Find the final balance and the dollar amount added each year.
  2. A \$500 account earns 8% compounded annually for 3 years. Find the final balance to the nearest cent.
  3. Use the Rule of 72 to estimate the doubling time at 8% annual growth. About how long would two doublings take?
  4. A \$6000 loan charges 12% APR compounded monthly and has a \$200 payment. Find the monthly rate, first-month interest, amount of principal reduction, and new balance after the payment.
  5. Loan A and Loan B both begin at \$5000 with the same APR. Loan A requires \$150 each month and Loan B requires \$250. Which has the same first-month interest? Which payment removes more principal in month 1?